Homework 3
STAT 132

1. Suppose that you observe a random sample, $X_1, \ldots, X_n$, from the normal distribution with mean μ and variance σ^2. Find the distribution of
$$\frac{n(\bar{X} - \mu)^2}{\sigma^2}.$$

2. Suppose that a point (X, Y, Z) is to be chosen at random in three-dimensional space, where X, Y, and Z are independent random variables, each following a standard normal distribution. What is the probability that the distance from the origin to the random point will be less than 1?

3. If a random variable X follows a χ^2 distribution with m degrees of freedom, then the distribution of $X^{1/2}$ is called a chi (χ) distribution with m degrees of freedom. Determine the mean of this chi distribution.

4. Suppose that X has the χ^2 distribution with 200 degrees of freedom. Explain why the central limit theorem can be used to determine the approximate value of $\mathbb{P}(160 < X < 240)$ and find this approximate value.

5. Let $X_1, \ldots, X_n$ be a random sample from the normal distribution with mean μ and variance σ^2. Also let $\widehat{\sigma^2}$ denote the sample variance. Find the smallest value of n for which the below expression is satisfied:
$$\mathbb{P}\left(\frac{\widehat{\sigma^2}}{\sigma^2} \leq 1.5\right) \geq 0.95$$